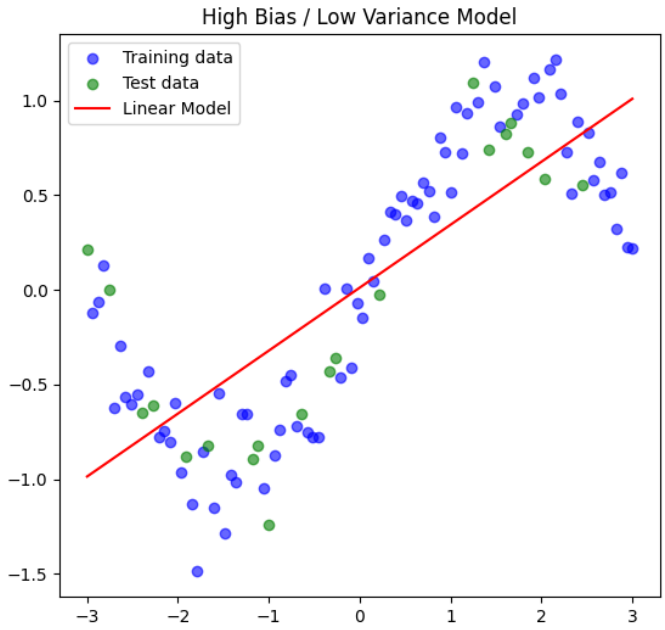
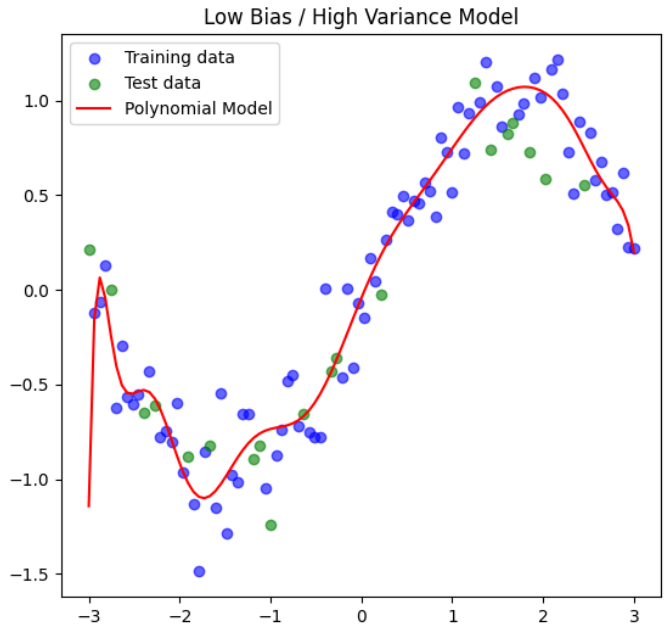
**High Bias / Low Variance and Low Bias / High Variance**




Linear Model Training MSE: 0.19743335485331392

Linear Model Testing MSE: 0.24695224333039673

Polynomial Model Training MSE: 0.029514806989774718

Polynomial Model Testing MSE: 0.1425297085836628

Linear Regression Model (High Bias/Low Variance)

Training MSE: 0.19743335485331392

Testing MSE: 0.24695224333039673

Polynomial Regression Model (Low Bias/High Variance)

Training MSE: 0.029514806989774718

Testing MSE: 0.1425297085836628

Bias - Variance of Linear Model

MSE: 0.253

Bias: 0.247

Variance: 0.006

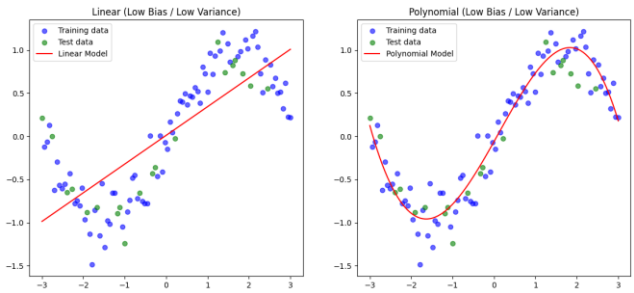
Bias - Variance of Polynomial Model

MSE: 2.609

Bias: 0.140

Variance: 2.468

## Low Bias / Low Variance




Linear Model Training MSE: 0.19743335485331392

Linear Model Testing MSE: 0.24695224333039673

Bias - Variance of Linear Model

MSE: 0.253

Bias: 0.247

Variance: 0.006

Polynomial Model Training MSE: 0.034392277465462945

Polynomial Model Testing MSE: 0.041525680224991667

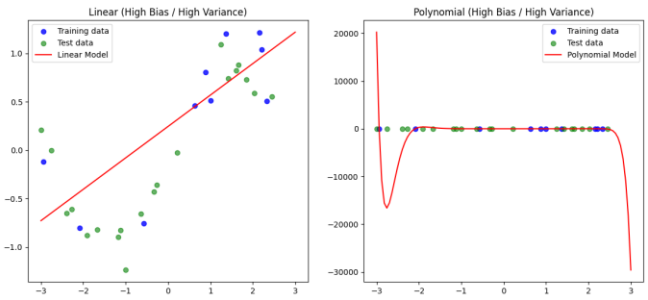
Bias - Variance of Polynomial Model

MSE: 0.045

Bias: 0.042

Variance: 0.003

## High Bias / High Variance




Linear Model Training MSE: 0.1811716081117703

Linear Model Testing MSE: 0.30139765378090844

Bias - Variance of Linear Model

MSE: 0.384

Bias: 0.292

Variance: 0.091

Polynomial Model Training MSE: 1.2074847170706594e-21

Polynomial Model Testing MSE: 35368838.221896544

Bias - Variance of Polynomial Model

MSE: 41793822606.081

Bias: 110291417.885

Variance: 41683531188.196